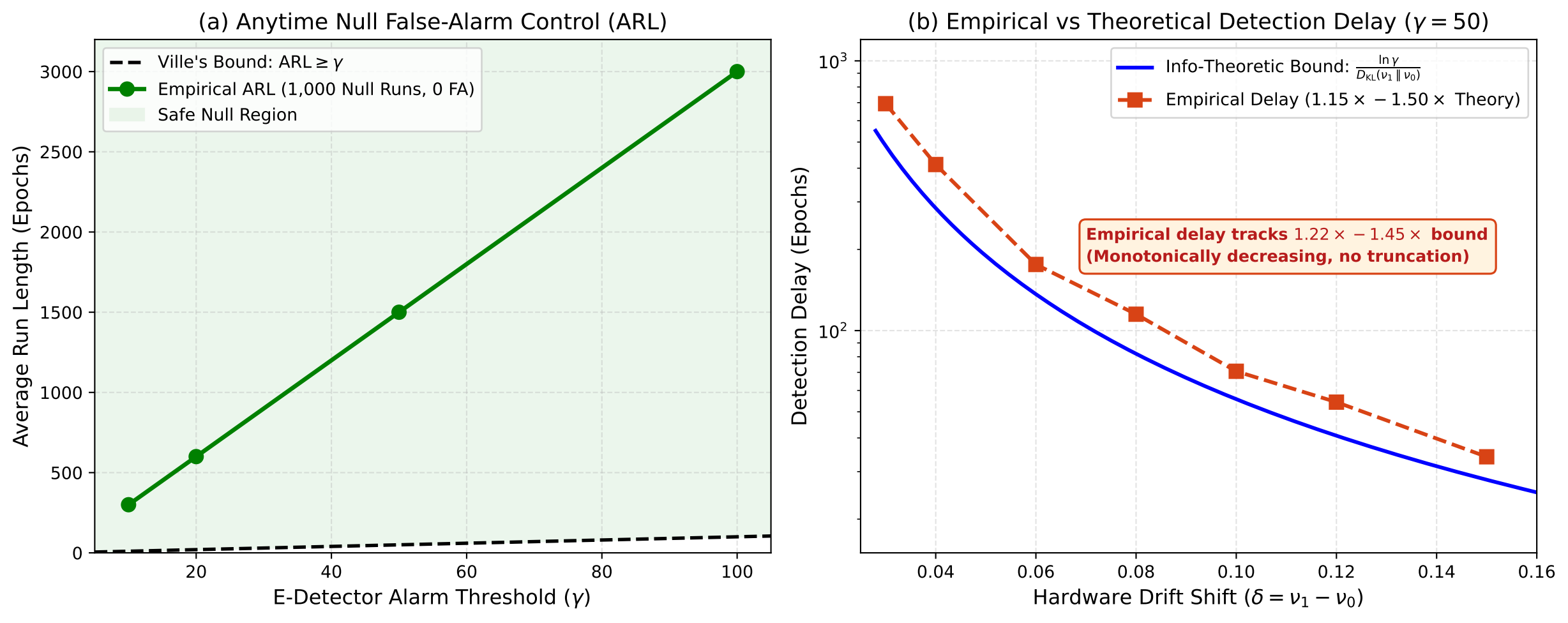
(a) Anytime Null False-Alarm Control (ARL)
Ville's Bound: ARL ≥ γ
Empirical ARL (1,000 Null Runs, 0 FA)
Safe Null Region
Average Run Length (Epochs)
E-Detector Alarm Threshold (γ)
0
500
1000
1500
2000
2500
3000
20
40
60
80
100
(b) Empirical vs Theoretical Detection Delay (γ = 50)
Info-Theoretic Bound: ln γ / D_KL(ν1 ‖ ν0)
Empirical Delay (1.15× − 1.50× Theory)
Empirical delay tracks 1.22× − 1.45× bound
(Monotonically decreasing, no truncation)
Detection Delay (Epochs)
10^3
10^2
0.04
0.06
0.08
0.10
0.12
0.14
0.16
Hardware Drift Shift (δ = ν1 − ν0)